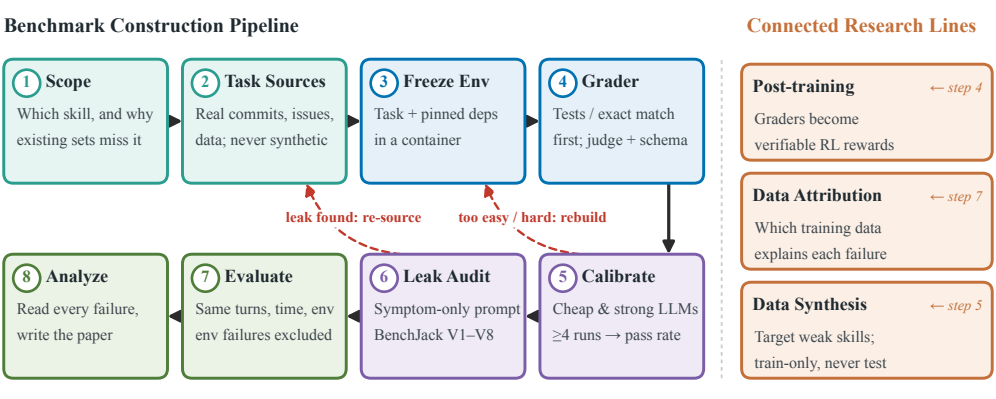

Benchmark Construction Pipeline
1 Scope
Which skill, and why existing sets miss it
2 Task Sources
Real commits, issues, data; never synthetic
3 Freeze Env
Task + pinned deps in a container
4 Grader
Tests / exact match first; judge + schema
leak found: re-source
too easy / hard: rebuild
8 Analyze
Read every failure, write the paper
7 Evaluate
Same turns, time, env env failures excluded
6 Leak Audit
Symptom-only prompt BenchJack V1–V8
5 Calibrate
Cheap & strong LLMs ≥4 runs → pass rate
Connected Research Lines
Post-training
← step 4
Graders become verifiable RL rewards
Data Attribution
← step 7
Which training data explains each failure
Data Synthesis
← step 5
Target weak skills; train-only, never test